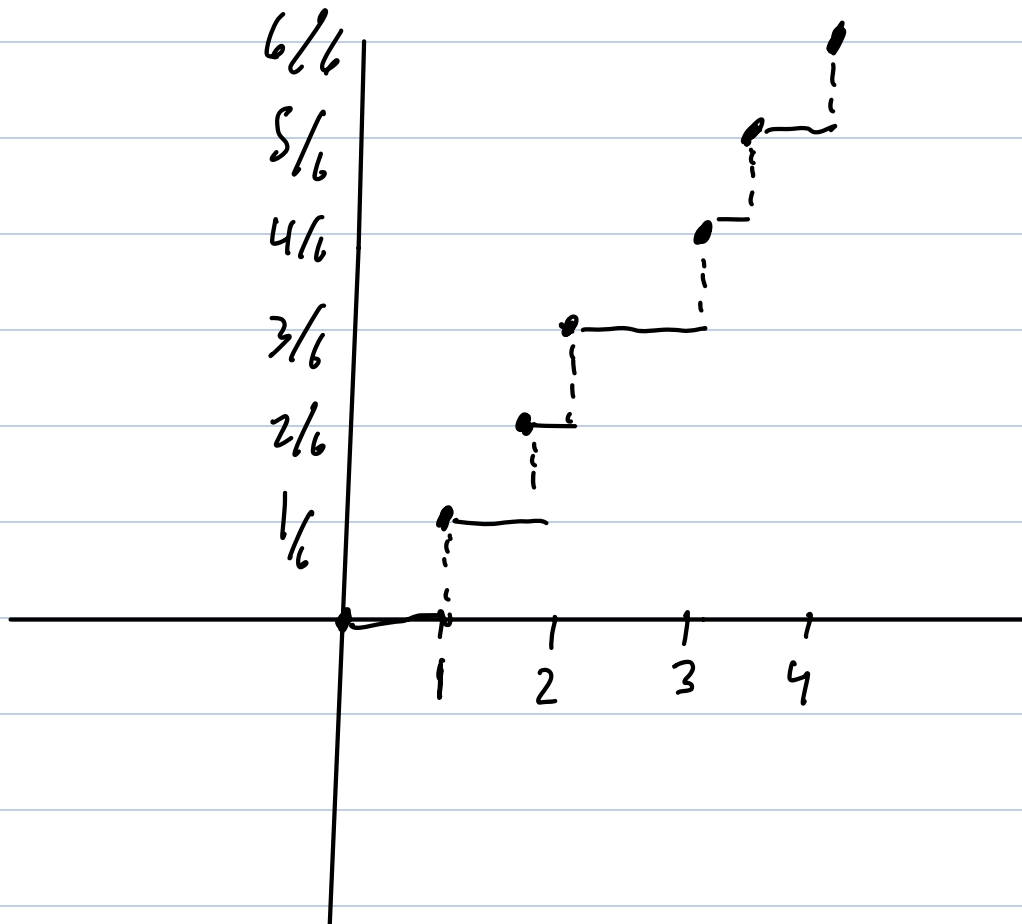


KDE Problems by Caleb Christian

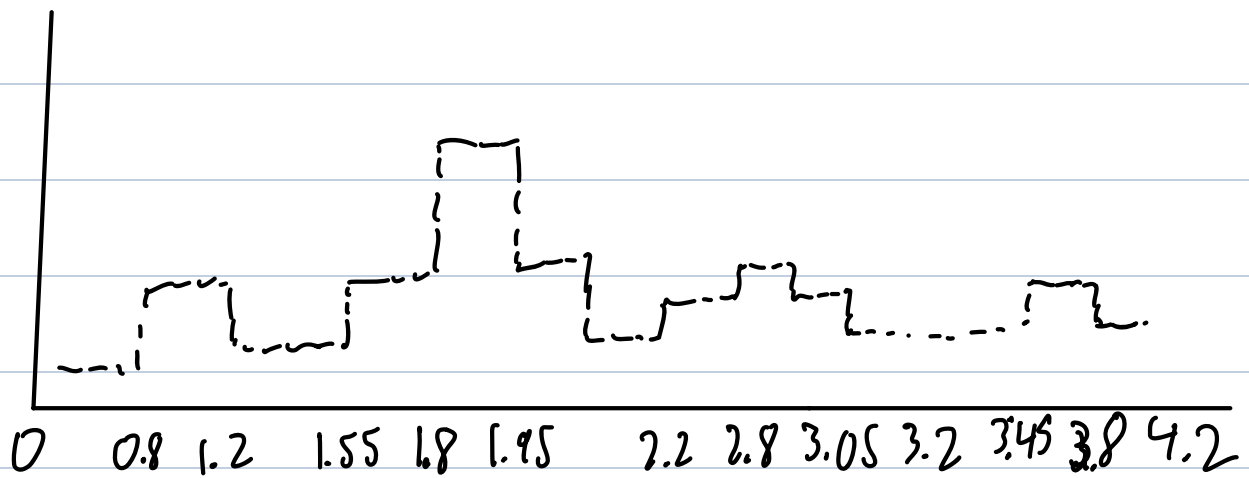
1) $[1, 1.75, 2, 3, 3.25, 4]$ $n=6$

a.)



b) $[0.8, 1.2], [1.85, 1.95], [1.8, 2.2],$
 $[2.8, 3.2], [3.05, 3.45], [3.8, 4.2]$

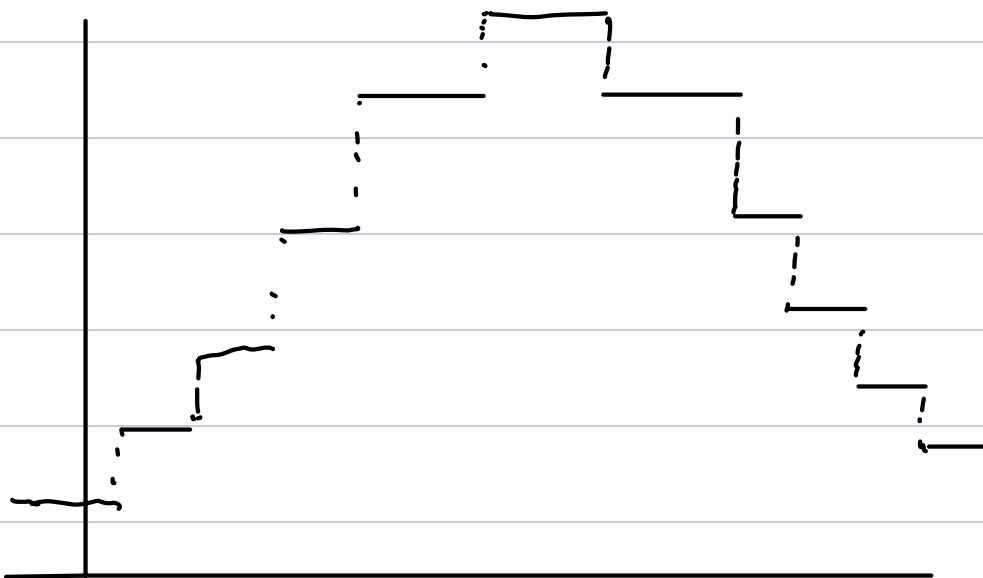
$$\frac{1}{2(0.2)(6)} = 0.417$$



c) we know $h = 1.84(\text{SD}) \cdot n^{-1/5}$, $n=6$

assuming SD is 1.095,
 $\therefore 1.84(1.095)(6)^{-1/5}$
 $= 1.406$

d) using the new bandwidth, we can adjust the original data: $[-0.406, 2.406]$, $[0.344, 3.156]$, $[0.594, 3.406]$, $[1.594, 4.406]$, $[1.844, 4.656]$, $[2.594, 5.406]$



e) There is more smoothness in the graph from d. The graph from b has more edges and shows more detail due to the smaller bandwidth.

$$2.) \quad x = [-1, -0.5, 0, 0.5, 1]$$

$$\begin{aligned} a) \quad \hat{f}_h(0) &= \frac{3}{2(5)(0.6)} \\ &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} b) \quad E(\hat{f}_h(0)) &= \frac{P(-0.6 < x < 0.6)}{2h} \\ &= \frac{P(x \leq 0.6) - P(x \leq -0.6)}{2(0.6)} \\ &= \frac{0.7257 - 0.2743}{1.2} \\ &= \frac{0.4514}{1.2} = 0.3762 \end{aligned}$$

$$c) \sin x = 0$$

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

$$f(0) = \frac{1}{\sqrt{2\pi}} e^{0/2}$$

$$= \frac{1}{\sqrt{2\pi}}$$

$$= 0.399$$

d) So as you can see, the two outputs from b & c are NOT equal. When it comes to bias, the KDE has a negative bias of about -0.02

$$e) h = 0.1$$

$$E[\hat{f}_h(0)] \approx \frac{P(-0.1 < X < 0.1)}{2(0.1)}$$

$$= \frac{P(X \leq 0.1) - P(X \leq -0.1)}{2(0.1)}$$

$$= \frac{0.5398 - 0.4602}{0.2}$$

$$= \frac{0.0796}{0.2} = 0.398$$

The gap is now even smaller. So as h decreases then the bias gets smaller.

f) I believe that $\hat{f}_h$ is a consistent estimator because when the bandwidth approaches 0, $\hat{f}_h$ or the KDE gets closer to the true density.

g) When choosing h , we have to make sure that the bandwidth is not too big which will lead to more bias, or not too small which will lead to more variance.

$$3) \quad x = [5, 7, 7, 8, 10], \quad h = 1.5$$

$$\begin{aligned} a). \quad \hat{f}_h &= \frac{\sum \{ |x_i - x| < h \}}{2nh} \\ &= \frac{\sum \{ |x_i - x| < h \}}{2(5)(1.5)} \\ &= \frac{\sum \{ |x_i - x| < h \}}{15} \end{aligned}$$

if $x = 6$, then $\hat{f}_h = 3/15$

if $x = 6.5$, then $\hat{f}_h = 2/15$

if $x = 7.0$, then $\hat{f}_h = 3/15$

if $x = 8.0$, then $\hat{f}_h = 3/15$

if $x = 8.5$, then $\hat{f}_h = 1/15$

b) gaussian: $\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x - x_i}{h} \right)^2}$

if $x = 6$, $\hat{f}_h = 0.1512$

if $x = 6.5$, $\hat{f}_h = 0.1687$

if $x = 7$, $\hat{f}_h = 0.1780$

if $x = 8$, $\hat{f}_h = 0.1674$

if $x = 8.5$, $\hat{f}_h = 0.1506$

Did all the work
on calculator, too
much space to take
up writing it down

c) so in part a, you can see the jump is from $x = 8$ to $x = 8.5$. It is probably due to its boundaries in the uniform kernel so if the observation lies outside the bandwidth, then it won't contribute to the data. The gaussian kernel has more of a smoother boundaries so the observations further out would contribute less GRADUALLY

d) Using c_j I would just choose gaussian kernel because of its smoother shape and there seems to be no gaps or jumps like the uniform kernel

4) c) I think when ejection-fraction is 20 to about 50 is when the original plot is reliable.

From 50 - 65 about, there seems to be a distinct variation. Also around when ejection-fraction is 80.